Ramanujan Class, Namakkal

Rank of a Matrix: Previous Year Questions

CSIR NET Mathematical Sciences · Problem Sheet

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Name: _____ **Date:** _____ **Score:** _____ / 30

Instructions. 30 questions. Some questions have more than one correct option; mark *all* correct options. Unless stated otherwise, matrices have real entries. Suggested time: 60 minutes.

Determinant, rank and idempotents

1. D is a non-zero $n \times n$ real matrix, $n \geq 2$. Which is true?

(A) If $\det D = 0$ then $\operatorname{rank} D = 0$
(B) If $\det D = 1$ then $\operatorname{rank} D \neq 1$
(C) If $\operatorname{rank} D = 1$ then $\det D \neq 0$
(D) If $\operatorname{rank} D = n$ then $\det D \neq 1$

2. A, B are $n \times n$ with $AB = BA = 0$ and $A + B$ invertible. Which are correct?

(A) $\operatorname{rank} A = \operatorname{rank} B$
(B) $\operatorname{rank} A + \operatorname{rank} B = n$
(C) $\operatorname{nullity} A + \operatorname{nullity} B = n$
(D) $A - B$ is invertible

3. A, B are $n \times n$ real matrices. Which is always true?

(A) $\operatorname{rank}(A + B) = \operatorname{rank} A + \operatorname{rank} B$
(B) $\operatorname{rank}(A + B) = \max(\operatorname{rank} A, \operatorname{rank} B)$
(C) $\operatorname{rank}(A + B) = \min(\operatorname{rank} A, \operatorname{rank} B)$
(D) $\operatorname{rank}(A + B) \leq \operatorname{rank} A + \operatorname{rank} B$

4. $n \geq 3$, J is the $n \times n$ matrix with every entry 1, and $A = I - \frac{1}{n}J$. Which is *not* true?

(A) $A^k = A$ for every $k \geq 1$
(B) $\operatorname{tr} A = n - 1$
(C) $\operatorname{rank} A + \operatorname{rank}(I - A) = n$
(D) A is invertible

5. A is $n \times n$ with $A^2 = A$. Which is *false*?

(A) eigenvalues of $I - A$ are 0 or 1
(B) $\operatorname{tr} A = \operatorname{rank} A$
(C) $\operatorname{rank} A + \operatorname{rank}(I - A) = n$
(D) every eigenvalue of A equals 1

Rank of products

6. P is 4×6, Q is 5×4, $\operatorname{rank} Q = 4$, $\operatorname{rank}(QP) = 2$. Then $\operatorname{rank} P =$

(A) 1 (B) 2 (C) 3 (D) 4

7. $m < n$, P is $n \times m$, Q is $m \times n$. Which is *not* possible?

(A) $\operatorname{rank}(PQ) = n$
(B) $\operatorname{rank}(QP) = m$
(C) $\operatorname{rank}(PQ) = m$
(D) $\operatorname{rank}(QP) = \left\lceil \frac{m+n}{2} \right\rceil$

8. A is 4×7, B is 7×4 and $AB = I_4$. Which are correct?

(A) $\operatorname{rank} A = 4$
(B) $\operatorname{nullity} B = 0$
(C) $BA = I_7$
(D) $\operatorname{rank} B = 7$

9. M is 7×5 of rank 3, N is 5×7 of rank 5. Then $\operatorname{rank}(MN) =$

(A) 5 (B) 3 (C) 2 (D) 1

10. A is a real 3×4 matrix of rank 2. Then $\operatorname{rank}(A^T A)$ is

(A) exactly 2 (B) exactly 3 (C) exactly 4 (D) at most 2, not nec. 2

11. [December 2023] Which of the following is *false*?

(A) The product of two 2×2 matrices of rank 2 has rank 2.
(B) The product of two 3×3 matrices of rank 2 has rank at most 2.
(C) The product of two 3×3 matrices of rank 2 has rank at least 2.
(D) The product of two 2×2 matrices of rank 1 can be the zero matrix.

12. $S = \{A \in M_3(\mathbb{R}) : A^T A = E_{11}\}$, where $E_{11} = \operatorname{diag}(1,0,0)$. Then S contains

(A) a nilpotent matrix (B) a matrix of rank one
(C) a matrix of rank two (D) a non-zero skew-symmetric matrix

Structured matrices

13. P is the 50×50 matrix with $p_{ij} = \min(i,j)$. Then $\operatorname{rank} P =$

(A) 1 (B) 2 (C) 25 (D) 50

14. J is the $n \times n$ all-ones matrix and $B = \begin{bmatrix} 0 & 0 & J \\ 0 & J & 0 \\ J & 0 & 0 \end{bmatrix}$ $(3n \times 3n)$. Then $\operatorname{rank} B =$

(A) 2 (B) $2n$ (C) $3n$ (D) 3

15. A is $n \times m$ with entries in $\{-1, 0, 1\}$; every column has exactly one 1 and exactly one -1. Then

(A) $\operatorname{rank} A \le n - 1$ (B) $\operatorname{rank} A = m$
(C) $n \le m$ (D) $n - 1 \le m$

16. a, b, c are distinct integers and $A = \begin{bmatrix} a & a^2 & a^5 \\ b & b^2 & b^5 \\ c & c^2 & c^5 \end{bmatrix}$. The set of all possible values of $\operatorname{rank} A$ is

(A) $\{3\}$ (B) $\{2, 3\}$ (C) $\{1, 2, 3\}$ (D) $\{0, 1, 2, 3\}$

17. $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & k+3 \end{bmatrix}$ has nullity 1. Then $k =$

(A) 1 (B) -1 (C) 0 (D) 2

18. X is 6×6 with 1 on the main diagonal, 2 on the anti-diagonal, 0 elsewhere. Then $\det X =$

(A) 27 (B) -27 (C) -9 (D) 81

19. $A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ and $f(x)$ is its minimal polynomial. Then $\operatorname{rank} f(A) =$

(A) 0 (B) 1 (C) 2 (D) 4

Rank, eigenvalues and forms

20. A, B are $n \times n$ real with $\det A > 0 > \det B$, and $C(t) = tA + (1-t)B$, $t \in [0,1]$. Then

(A) $C(t)$ is invertible for each $t \in [0,1]$
(B) there is $t_0 \in (0,1)$ with $C(t_0)$ not invertible
(C) $C(t)$ is not invertible for each $t \in [0,1]$
(D) $C(t)$ is invertible for only finitely many $t \in [0,1]$

21. A is $n \times n$ and has exactly r distinct non-zero (complex) eigenvalues. Then

(A) $\operatorname{rank} A \le r$ (B) $\operatorname{rank} A = r$
(C) $\operatorname{rank} A \ge r$ (D) A^2 has exactly r distinct non-zero eigenvalues

22. A is the real symmetric matrix with $(x,y,z)A(x,y,z)^T = xy - z^2$. Let p, q be the numbers of positive and negative eigenvalues of A. Then

(A) $p = 1$ (B) $p = 2$ (C) $q = 2$ (D) $\operatorname{rank} A = 2$

23. $\langle \cdot, \cdot \rangle$ is a symmetric bilinear form on $\mathbb{R}^2$ with matrix A, and $\langle v, v \rangle > 0$, $\langle w, w \rangle < 0$ for some v, w. Then

(A) $A^2 = 0$ (B) $\operatorname{rank} A = 1$
(C) $\det A > 0$ (D) $\langle u, u \rangle = 0$ for some $u \neq 0$

24. M is $n \times n$ Hermitian of rank k; $\lambda \neq 0$ is an eigenvalue with unit eigenvector u. Then

(A) $\operatorname{rank}(M - \lambda uu^*) = k - 1$ (B) $\operatorname{rank}(M - \lambda uu^*) = k$
(C) $\operatorname{rank}(M - \lambda uu^*) = k + 1$ (D) $(M - \lambda uu^*)^2 = M^2 - \lambda^2 uu^*$

Subspaces, RREF and submatrices

25. A is $n \times n$; its first three rows are linearly independent and its first five columns are linearly independent. Then

(A) $n \ge 5$ (B) $\operatorname{rank} A = 5$
(C) $\operatorname{rank} A \ge 5$ (D) $\operatorname{rank}(A^2) \ge 5$

26. A is $m \times n$; $A_{(1,:)}$, $A_{(:,1)}$, $A_{(1,1)}$ delete row 1, column 1, both. Let r, r_1, r_2, r_{11} be their ranks. Then

(A) $r - 2 \le r_{11} \le r$ (B) $r_1 = r_2$
(C) if $r = r_1 = r_2$ then $r_{11} = r$ (D) $r_1 + r_2 \ge 2r - 2$

27. (I) Some 4×4 real matrix has row space with basis $\{(1,0,0,1), (0,1,1,0)\}$ and nullity 1.
(II) Some 4×4 real matrix has column space with basis $\{(1,0,0,1)^T, (0,1,1,0)^T\}$ and nullity 2.

(A) (I) false, (II) true (B) (I) true, (II) false
(C) both true (D) both false

28. M is 5×5 of rank 3, R is its row-reduced echelon form and r_i the i-th row of R. Which is possible?

(A) $r_3 = (0,1,0,0,0)$ (B) $r_5 = (0,0,0,0,1)$
(C) $r_4 = (0,0,0,1,0)$ (D) $r_4 = (0,0,0,0,0)$

29. A is 4×4 with null space $N(A) = \{(x,y,z,w) : x+y+z=0,\ y+w=0\}$. Which are correct?

(A) $\operatorname{rank} A = 2$ (B) $\operatorname{nullity} A = 3$
(C) $\{(1,0,-1,0),(1,-1,0,1)\}$ is a basis of $N(A)$
(D) $\{(1,1,1,1),(0,0,0,1)\}$ is a basis of $N(A)$

30. [2023] Which of the following is correct?

(A) There are real A (3×2) and B (2×3) with $AB = I_3$.
(B) For $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ there is an integer matrix B with $AB = I_2$.
(C) For $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ over $\mathbb{Z}_6$ there is B over $\mathbb{Z}_6$ with $AB = I_2$.
(D) For every non-zero 3×3 real diagonal A there is B with $AB = I_3$.

Practice daily. Best wishes for CSIR NET!